

SE LAB 1

PROBLEM STATEMENT 15 - Healthcare & Telemedicine

Pharmacy Expiry & Re-order Dispatch Engine

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- Complete Requirements Table:

1. Functional Requirements (FR)

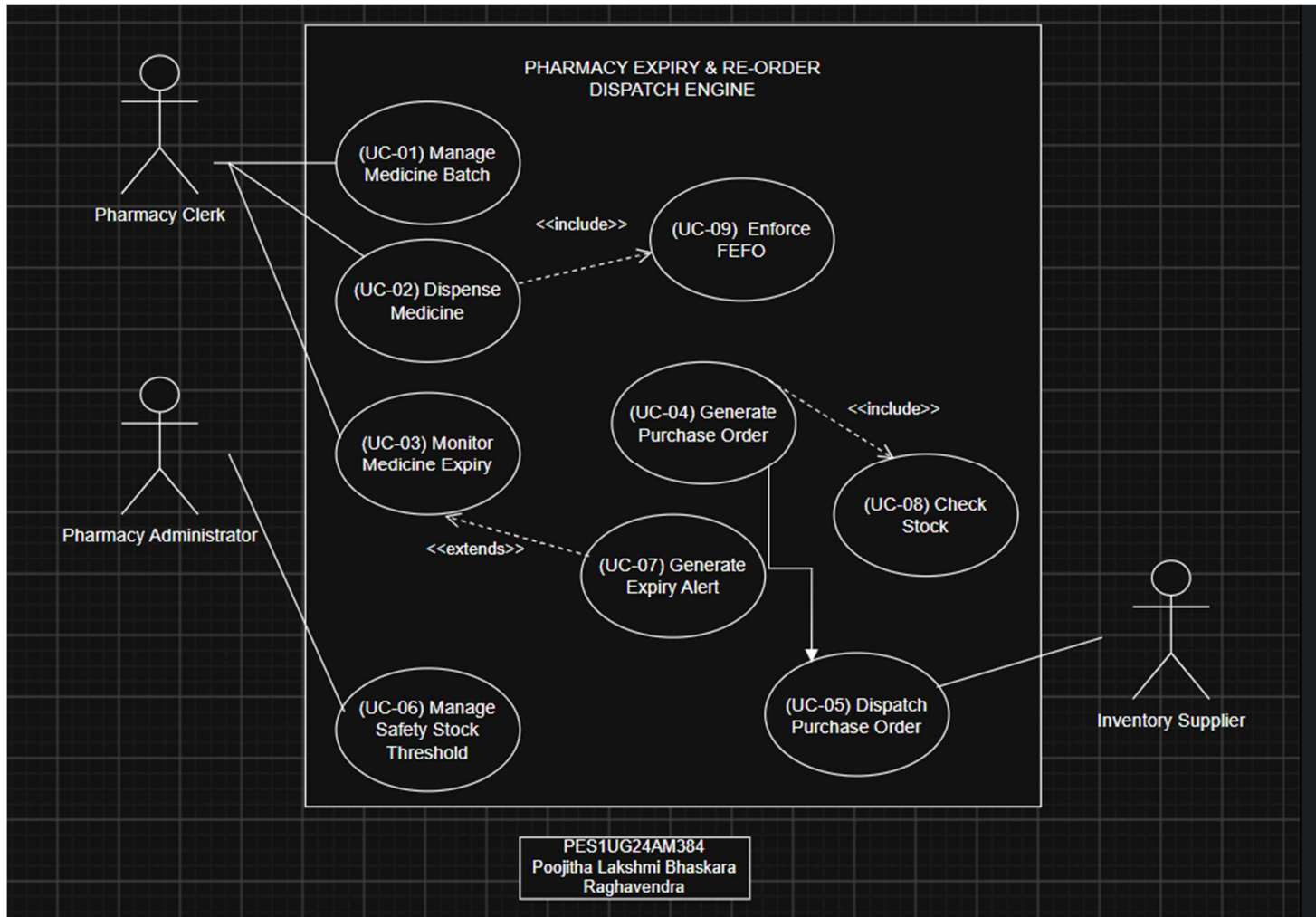
Req ID	Type	Description	Priority	Acceptance Criteria	Rationale
FR-001	Functional	The system shall enforce FEFO picking order by prioritizing medicine batches with the nearest expiry date during sales checkout.	High	Pass: Oldest valid batch selected first; Fail: Newer batch assigned while older unexpired batch exists.	Prevents medicine wastage due to expiry and ensures proper stock rotation.
FR-002	Functional	The system shall allow the Pharmacy Clerk to record and update medicine batch details, including medicine name, batch number, quantity, manufacturing date, and expiry date.	High	Pass: All valid batch details are stored and retrievable. Fail: The system saves incomplete or invalid batch information.	Accurate batch information is required for expiry tracking and dispensing.
FR-003	Functional	The system shall automatically identify and flag medicine batches approaching their expiry date.	High	Pass: Batches within the configured expiry-warning period are displayed as alerts. Fail: An eligible near-expiry batch is not flagged.	Helps pharmacy staff take action before medicines expire.
FR-004	Functional	The system shall automatically generate a purchase order request when the available quantity of a medicine falls below its predefined safety-stock threshold.	High	Pass: A purchase order is generated when stock falls below the threshold. Fail: No	Prevents stock-outs of essential medicines.

				order is generated when the threshold condition is satisfied.	
FR-005	Functional	The system shall dispatch generated purchase orders to the Inventory Supplier and record the order status.	Medium	Pass: The supplier receives the purchase order and the system records its status as dispatched. Fail: The order is generated but not dispatched or recorded.	Ensures that re-order requests reach the supplier and can be tracked.

2. Non-Functional Requirements (NFR)

Req ID	Type	Description	Priority	Acceptance Criteria	Rationale
NFR-001	Nonfunctional	The system shall automatically generate and dispatch supplier purchase-order requests when item stock falls below the safety threshold while maintaining the required performance and security standards.	High	Pass: Benchmarking tests confirm the required response time and security standards under simulated peak load. Fail: The system exceeds the defined response limit or violates security requirements.	Ensures automated re-ordering remains reliable during high pharmacy activity.
NFR-002	Nonfunctional	The system shall ensure that only authorized Pharmacy Clerks can modify medicine inventory and batch information.	High	Pass: Unauthorized users are denied access to inventory modification functions. Fail: An unauthorized user can add, edit, or delete inventory data.	Protects critical pharmacy inventory data from unauthorized changes.

- UML Use-Case Diagram:



- Use-Case Flow Specification:

Primary Actor : Pharmacy Clerk

Use Cases :

- 1) Manage Medicine Batch
- 2) Dispense Medicine
- 3) Monitor Medicine Expiry
- 4) Generate Purchase Order
- 5) Dispatch Purchase Order
- 6) Manage Safety Stock Threshold
- 7) Generate Expiry Alert
- 8) Check Stock

Use Case: Dispense Medicine

Goal: Dispense the requested medicine using the appropriate batch while enforcing FEFO.

Preconditions :

1. The Pharmacy Clerk is authorized to access the inventory system.
2. The requested medicine is registered in the pharmacy inventory.
3. At least one valid, unexpired batch of the requested medicine is available.

Postconditions :

1. The requested medicine quantity is deducted from the selected batch.
2. The inventory quantity is updated.
3. The dispensing transaction is recorded.
4. The batch with the nearest valid expiry date is used first.

Main Success Scenario :

1. Pharmacy Clerk selects the "Dispense Medicine" option.
2. System displays a form to enter the medicine name and required quantity.
3. Pharmacy Clerk enters the medicine name and required quantity.
4. System validates the entered medicine and quantity.
5. System retrieves all available batches of the requested medicine.
6. System filters out batches that have already expired or have no available stock.
7. System applies FEFO and identifies the valid batch with the nearest expiry date.

8. System displays the selected batch and quantity to the Pharmacy Clerk.
9. Pharmacy Clerk confirms the dispensing.
10. System deducts the dispensed quantity from the selected batch.
11. System updates the medicine inventory.
12. System records the dispensing transaction.
13. System displays a successful dispensing confirmation.
14. Use case ends successfully.

Alternate Flow :

7a. No Valid Batch Available

1. System finds that all available batches are either expired or out of stock.
2. System displays a "Medicine unavailable" message.
3. System does not complete the dispensing transaction.
4. Pharmacy Clerk may search for another medicine or cancel the operation.
5. Use case ends.

7b. Requested Quantity Exceeds Selected Batch

1. System identifies that the requested quantity is greater than the quantity available in the earliest-expiring batch.
2. System checks the next valid batch according to FEFO order.
3. System combines quantities from subsequent valid batches if sufficient stock exists.
4. System updates the quantities of the batches used.
5. If sufficient stock does not exist, the system displays "Insufficient stock" and the dispensing operation is cancelled.